

NuMicro[®] Family VSCode Quick Start Guide

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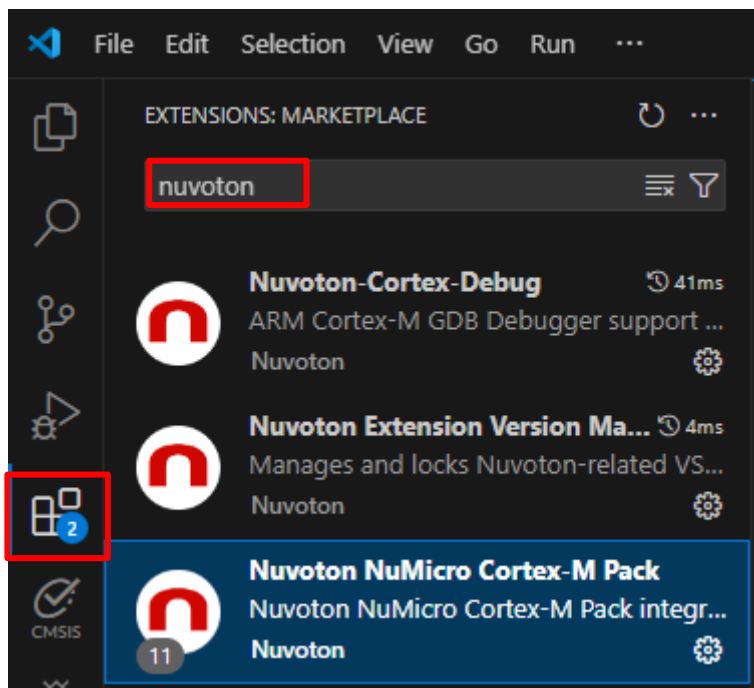
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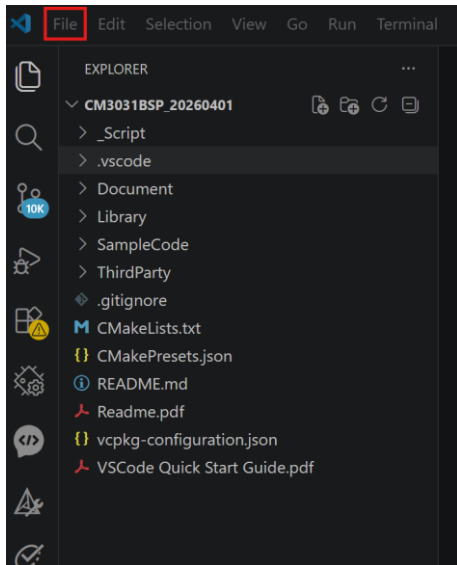
1 INSTALLATION VSCODE AND EXTENSIONS

1. Download VSCode from <https://code.visualstudio.com> and install it.
2. Launch VSCode and click Extensions in the Activity Bar.
3. Text “**Nuvoton NuMicro Cortex-M Pack Extension**” in search bar. Click install it.

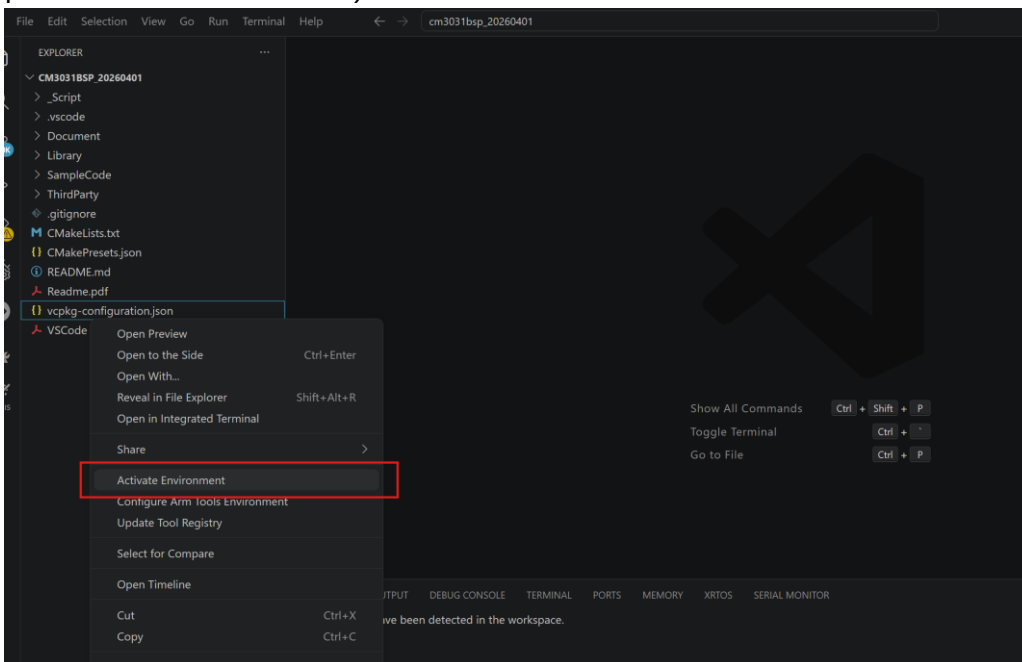


2 GET STARTED WITH AN EXAMPLE PROJECT

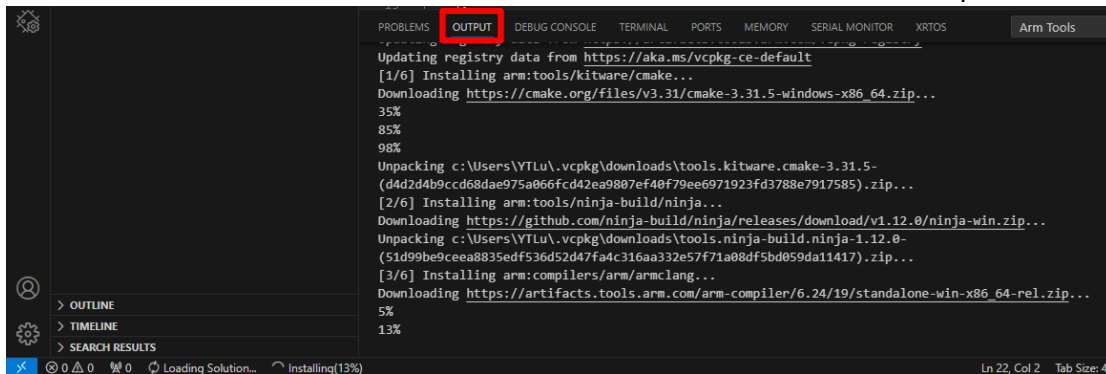
1. Click **File** and select "**Open Folder**" in the toolbar. Then select the path of the example project as below.



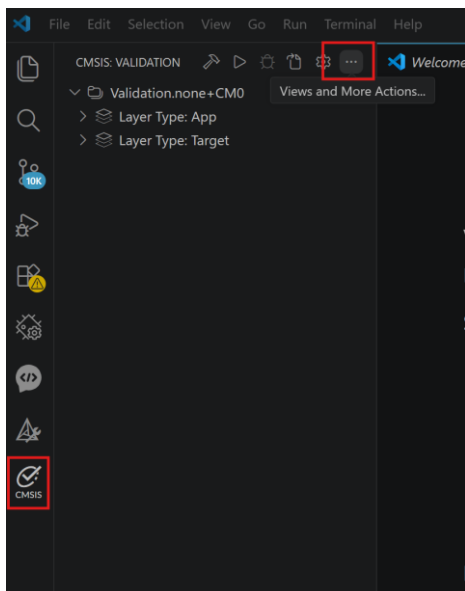
2. Right-click `vcpkg-configuration.json` and select **Activate Environment**. (Please use private network connection)



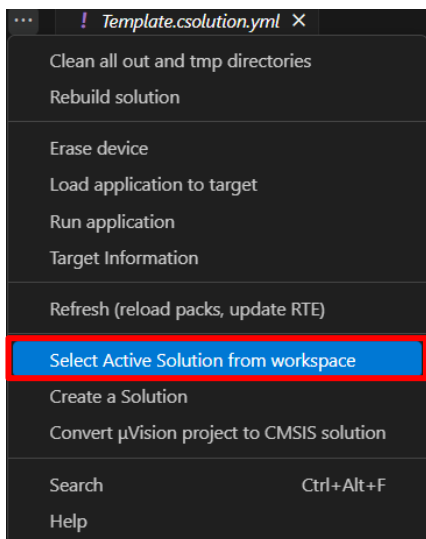
3. Check OUTPUT terminal at the bottom. It will download and install the requires tools.



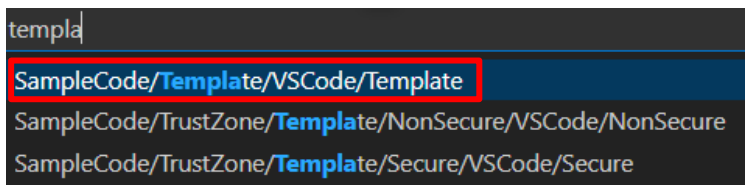
4. Click CMSIS in the Activity Bar.
Then click “Views and More Actions” button.



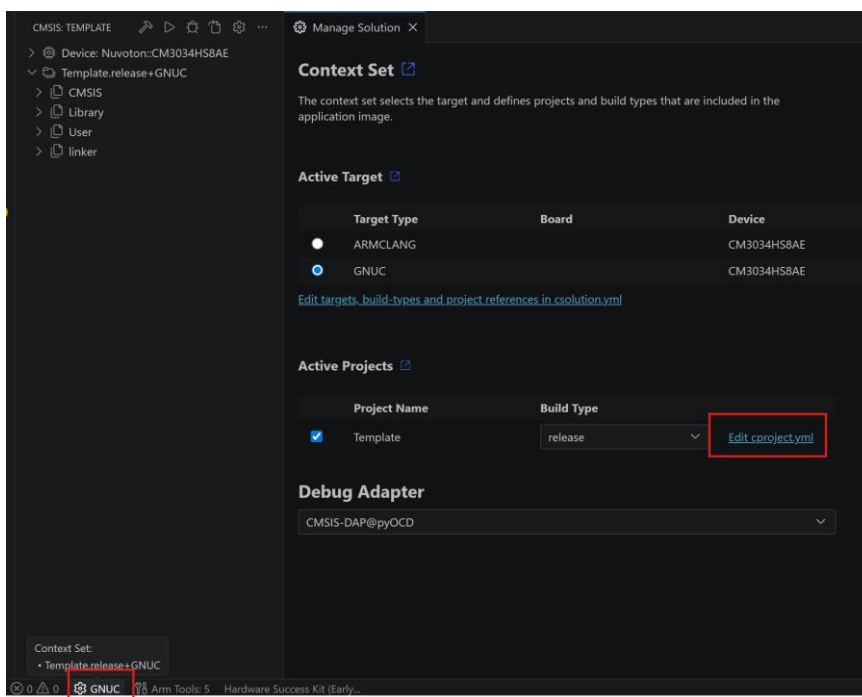
5. Select Active solution from workspace.



6. Select an example code to active it.



7. Click "Context Set", and then click "Edit cproject.yml" to open project settings.



Add setting parameters in cproject.yml.

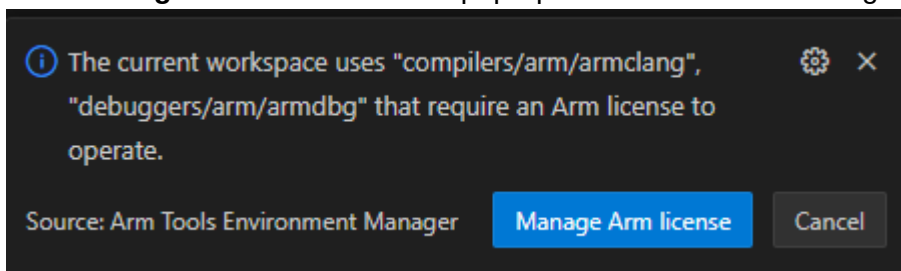
```

1 project:
2   packs:
3     - pack: Nuvoton::NuMicroCM33_DFP
4   misc:
5     - for-compiler: AC6
6     C: [-std=c99, -fno-rtti, -funsigned-char, -fshort-enums, -fshort-wchar, -ffunction-sections, -fdfat
7     ASM: [-D_MICROLIB]
8     Link: [--library_type=microlib, --map, --load_addr_map_info, --symbols, --info sizes, --info total
9     -for-compiler: GCC
10    C: [-std=gnu11, --specs=nano.specs, -Wno-return-type, -Wno-unused-but-set-variable, -fdata-section
11    Link:
12      - -Xlinker
13      - --gc-sections
14      - --specs=nano.specs
15      - --specs=nosys.specs
16      - -Wl,-Map,"$${OutDir}/$${Project$.map}"
17
18 setups:
19   - output:
20     | type: ["elf", "bin"]
21 linker:
22   - for-compiler: GCC
23     script: ../../../../Library/Device/Nuvoton/cm3031/Config/cm3031_gcc.ld
24 add-path:
25   - ../../../../Library/Device/Nuvoton/cm3031/Include
26   - ../../../../Library/StdDriver/inc
27   - ../../../../Library/CMSIS/Core/Include
28 groups:

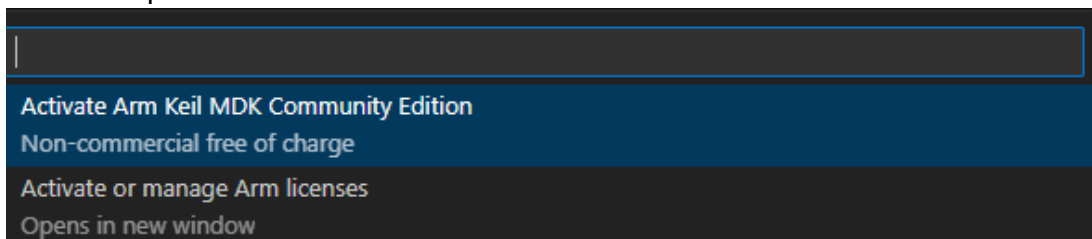
```

3 MANAGE ARM LICENSE

1. Click **Manage Arm License** in the pop-up window at the bottom right.



2. In the search bar at the top, select **"Activate or manage Arm licenses"** from the available options.

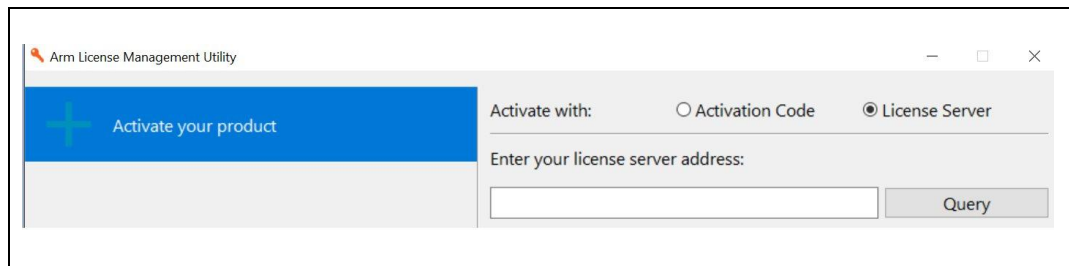


3. Get Keil MDK License ID code
 - a. Navigation to Official Website



- b. Fill Out the Form

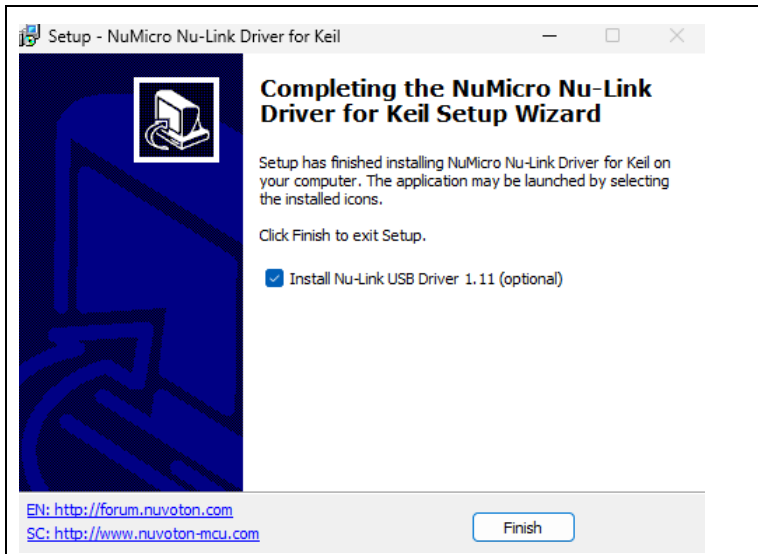
c. Check Mailbox and Fill in the License Server



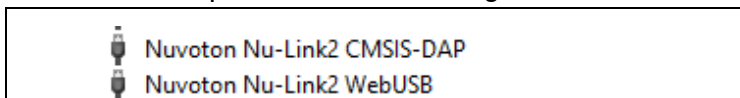
The screenshot shows a window titled "Arm License Management Utility". On the left, there is a blue button with a green plus icon and the text "Activate your product". On the right, there are two radio buttons under the heading "Activate with:". The first is "Activation Code" and the second is "License Server", which is selected. Below this, there is a text input field labeled "Enter your license server address:" and a "Query" button.

4 CONFIGURE THE DEVICE

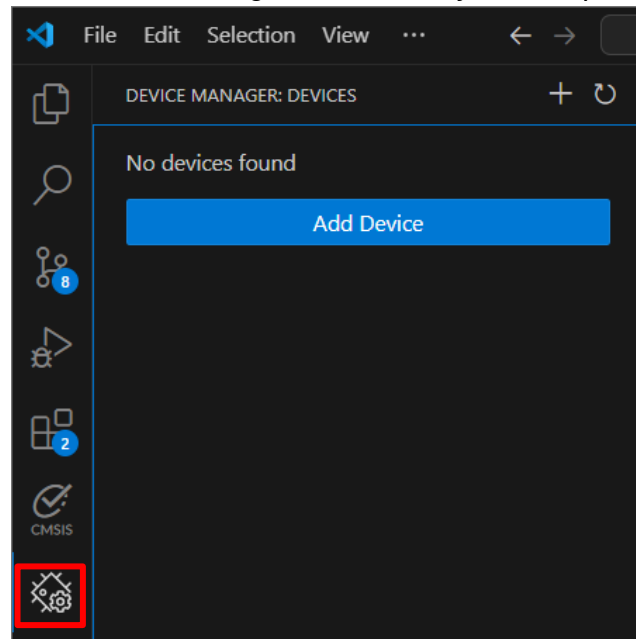
1. Install Nuvoton Nu-Link Keil Driver
2. After installing the Keil driver, please check the box to install the Nu-Link USB Driver.



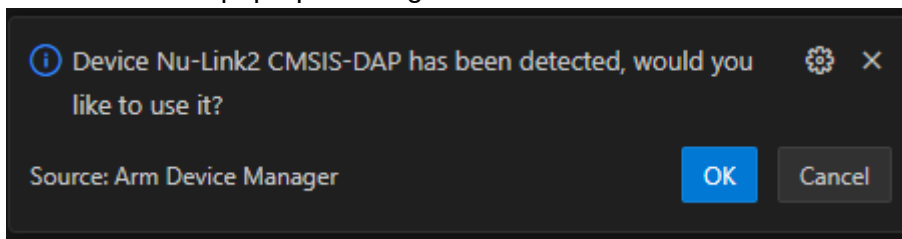
3. Installation complete in Device Manager



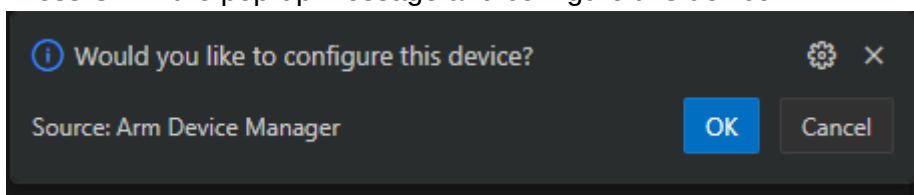
4. Click Device Manager in the Activity Bar to open the Device Manager.



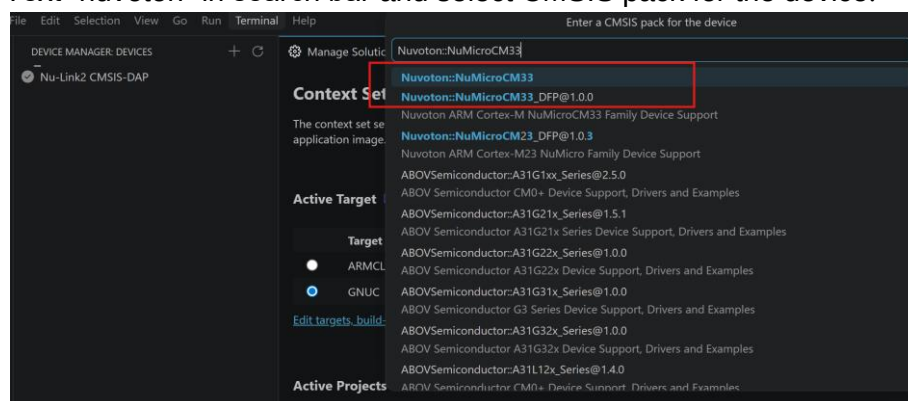
5. Connect device to your computer over USB.
The Device Manager detects the board and displays a pop-up message.
Press OK in the pop-up message and use it.



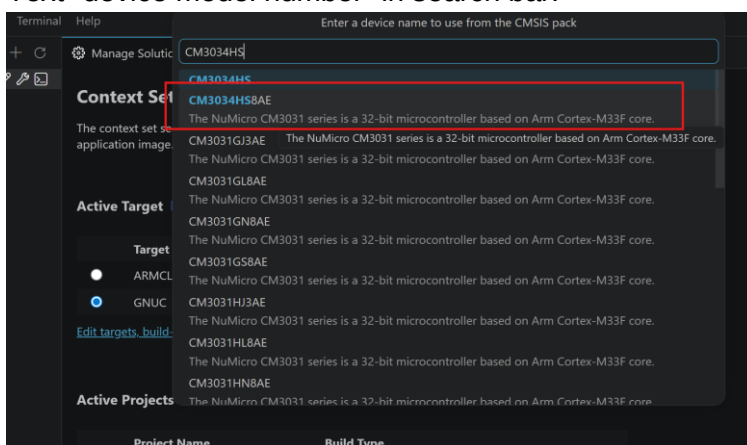
6. Press OK in the pop-up message and configure this device.



7. Text “nuvoton” in search bar and select CMSIS pack for the device.

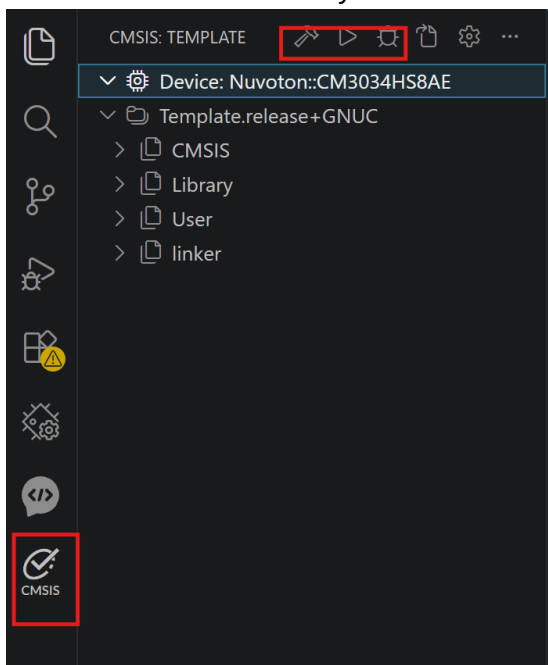


8. Text “device model number” in search bar.



5 RUN THE EXAMPLE PROJECT

1. Click CMSIS in the Activity Bar



2. Click Build



```
Execute: cbuild d:\CM3031\BSP\cm3031bsp_20260401\SampleCode\Template\VSCode\Template.csolution.yml --context-set --packs
+-----+
(1/1) Building context: "Template.release+GNUC"
Using GCC V14.3.1 compiler, from: 'c:/Users/XDMo/.vcpkg/artifacts/2139c4c6/compilers.arm.arm.none.eabi.gcc/14.3.1/bin'
Building CMake target 'Template.release+GNUC'
ninja: no work to do.
+-----+
Build summary: 1 succeeded, 0 failed - Time Elapsed: 00:00:05
+-----+
Completed: cbuild succeed with exit code 0
Build complete
[*] Terminal will be reused by tasks, press any key to close it.
```

3. Click Load & Run

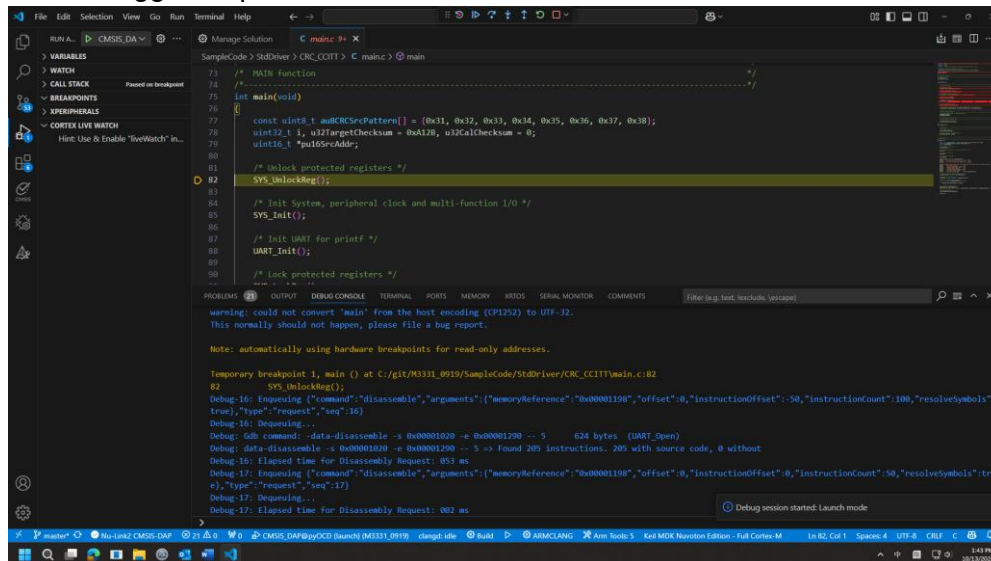


```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS MEMORY XRTOS SERIAL MONITOR
[*] Executing task: pyocd load --probe cmsisdap: --cbuild-run d:\CM3031\BSP\cm3031bsp_20260401\SampleCode\Template\VSCode\Template
File "threading.py", line 1045, in _bootstrap_inner
File "pyocd\debug\svd\loader.py", line 72, in run
File "pyocd\debug\svd\loader.py", line 48, in load
File "pyocd\debug\svd\parser.py", line 74, in for_xml_file
File "xml\etree\ElementTree.py", line 1219, in parse
0001187 I Loading D:\CM3031\BSP\cm3031bsp_20260401\SampleCode\Template\VSCode\out\Template\GNUC\release\Template.elf [load_cmd]
File "xml\etree\ElementTree.py", line 570, in parse
TypeError: expected str, bytes or os.PathLike object, not NoneType
[=====] 100%
0001797 I Erased 9216 bytes (9 sectors), programmed 9216 bytes (9 pages), skipped 0 bytes (0 pages) at 14.82 kB/s [loader]
[*] Terminal will be reused by tasks, press any key to close it.
```

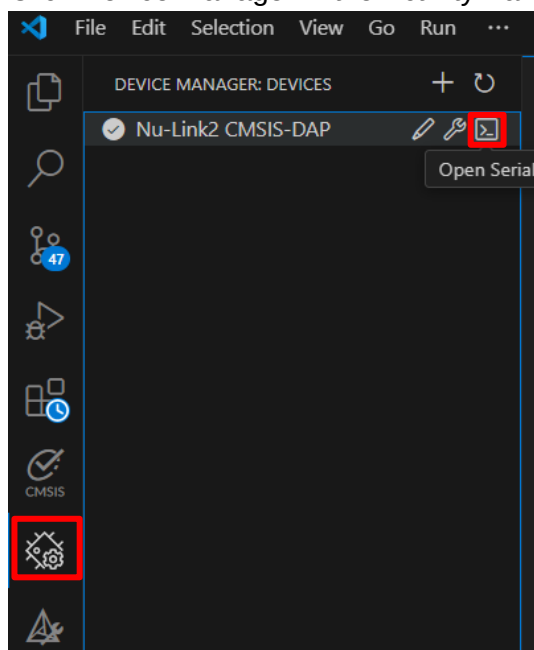
4. Click Load & Debug



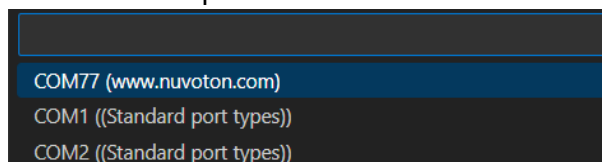
The debugger stops at the main function.



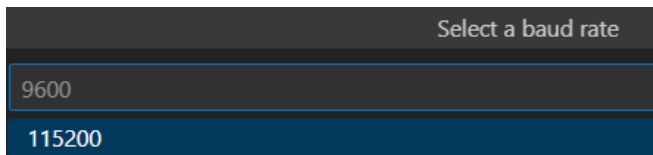
5. Click Device Manager in the Activity Bar and open serial port.



Select a serial port: COM77



Select a baud rate: 115200



Display Terminal window as below

```

PROBLEMS 21 OUTPUT DEBUG CONSOLE TERMINAL PORTS MEMORY XRTOS SERIAL MONITOR CC
Opened with baud rate: 115200

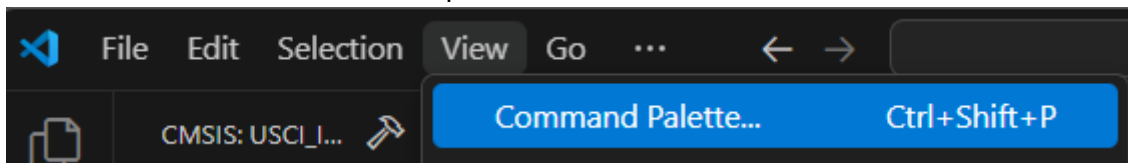
CPU @ 160000000 Hz
+-----+
| CRC-CCITT Polynomial Mode Sample Code |
+-----+

# Calculate [0x31, 0x32, 0x33, 0x34, 0x35, 0x36, 0x37, 0x38] CRC-CCITT checksum value.
- Seed value is 0xFFFF
- CPU write data length is 16-bit
- Checksum complement disable
- Checksum reverse disable
- Write data complement disable
- Write data reverse disable
- Checksum should be 0xA12B

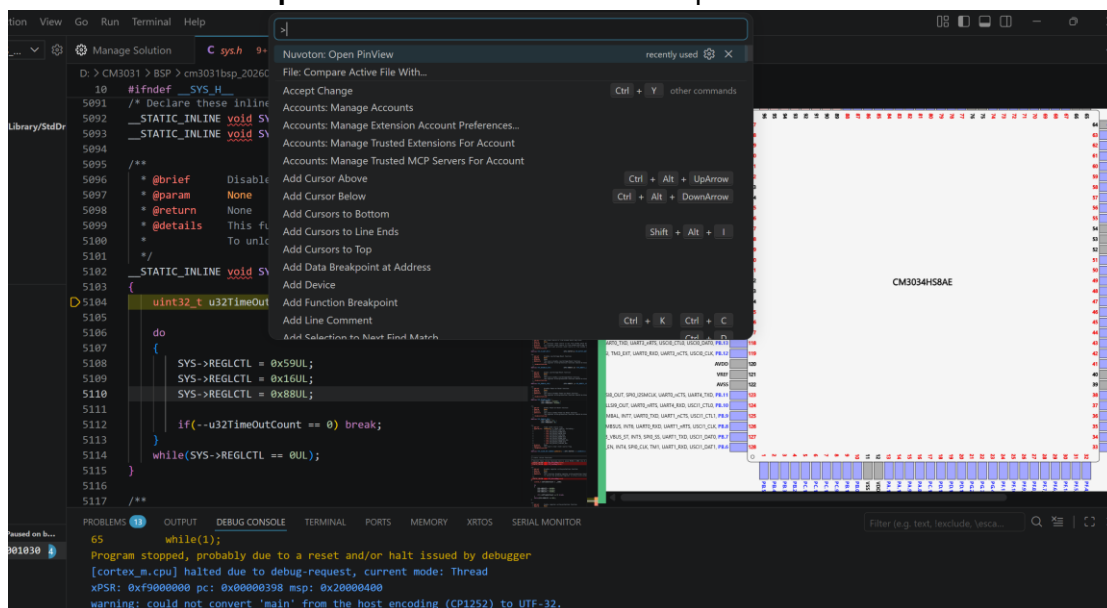
CRC checksum is 0xA12B ... PASS.

```

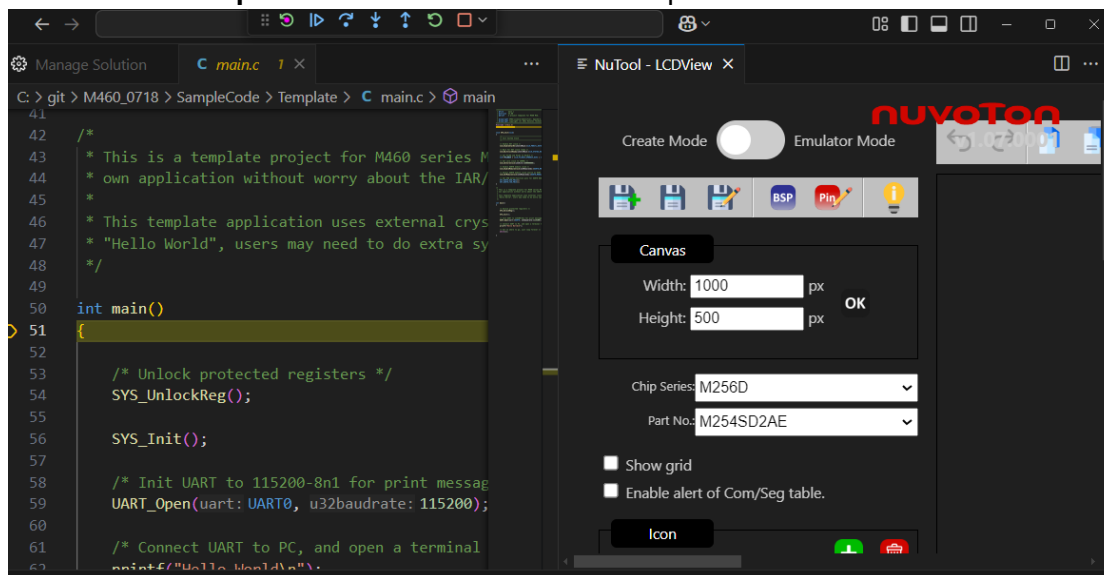
- Click View and select “Command palette” in the toolbar.



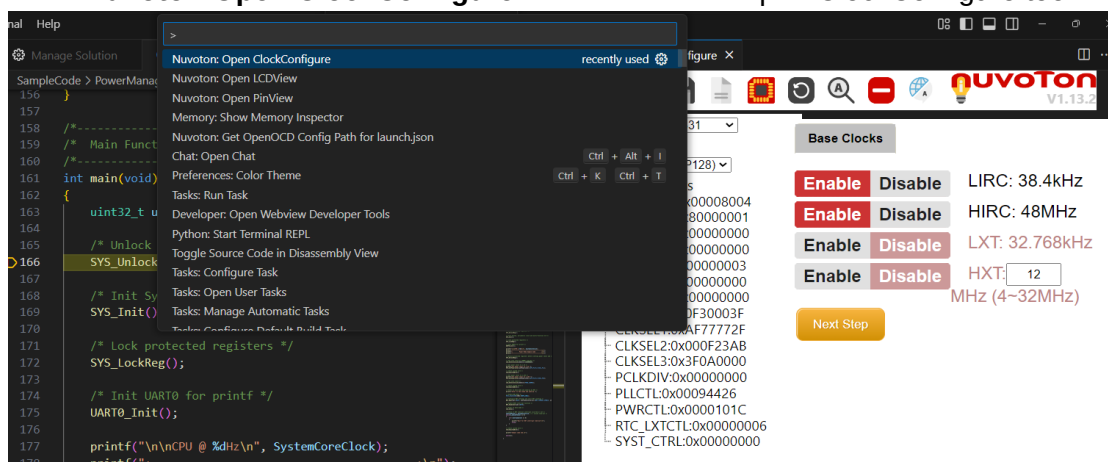
- Text “Nuvoton:Open PinView” in search bar to open PinView tool



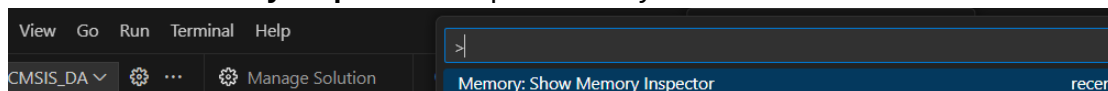
- Text “Nuvoton:Open LCDView” in search bar to open PinLCD tool.



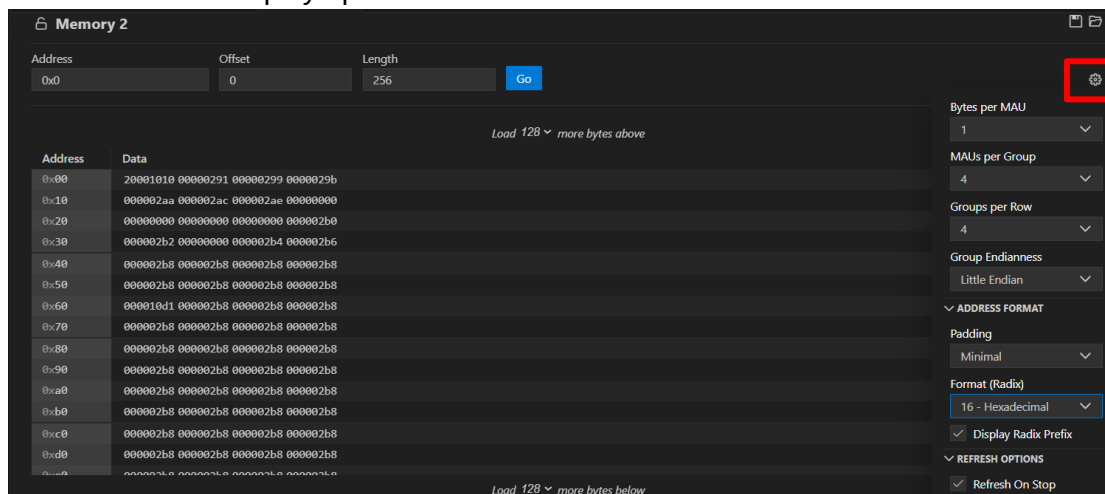
- Text “Nuvoton:Open ClockConfigure” in search bar to open ClockConfigure tool.



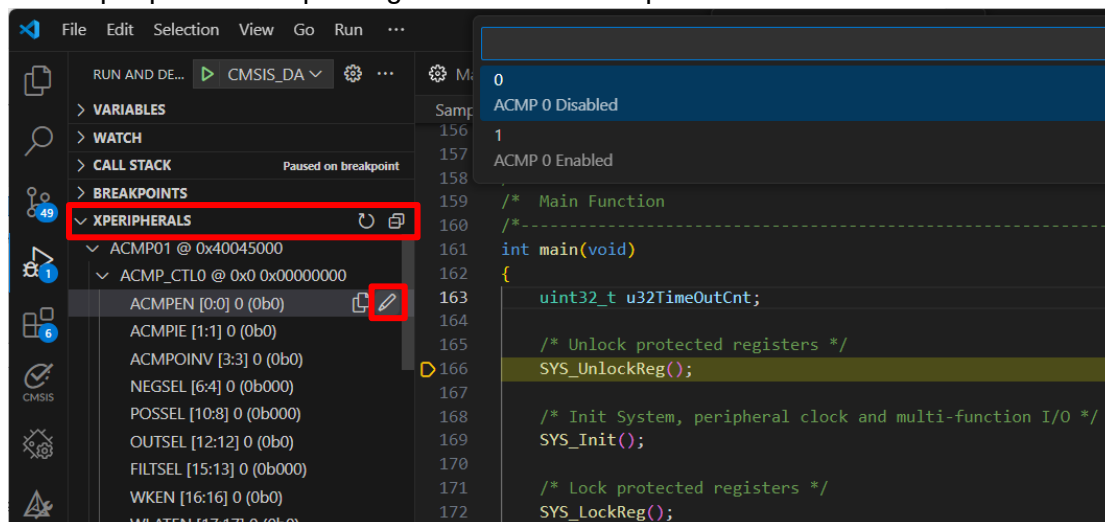
- Text “Show Memory Inspector” to open memory window.



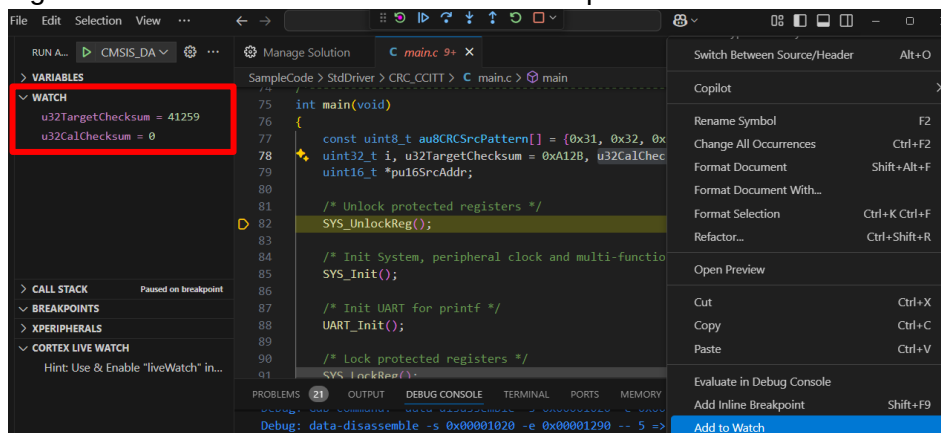
Click “advanced display options” button to set address format.



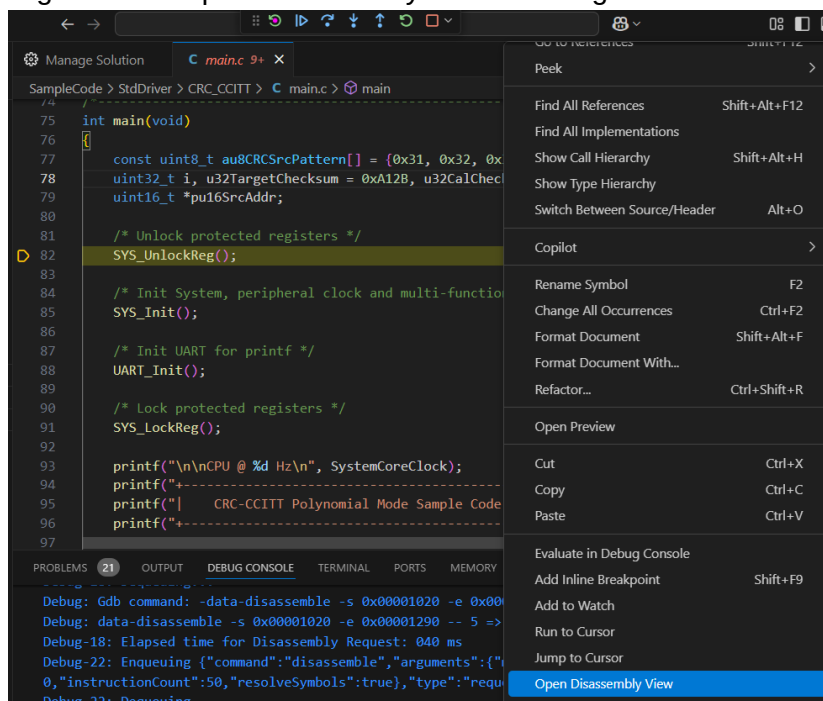
11. Click Xperipherals to open register window and update value.



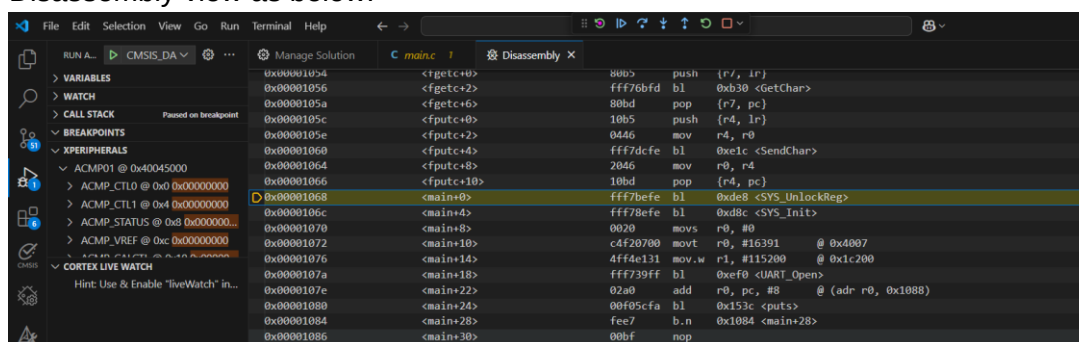
12. Right-click on variable u32CalChecksum to open add to watch.



13. Right-click to open disassembly view in debug mode.



14. Disassembly view as below.



REVISION HISTORY

Date	Revision	Description
2026.4.2	1.00	1. Initially version.

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